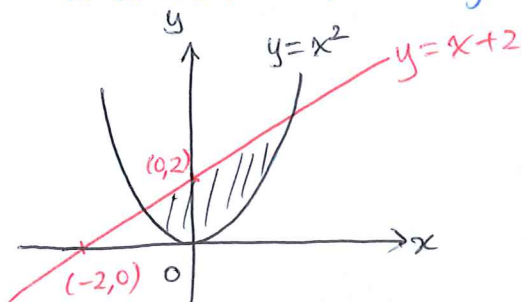


Week 8 NOTES

Prepared by Hugo MAK

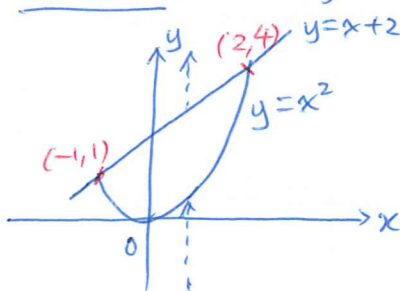
Reference Chapters: Adams and Essex Ch. 14.4, 14.5, 14.6

Larson and Edwards Ch. 14.3, 14.6, 14.7, 14.8

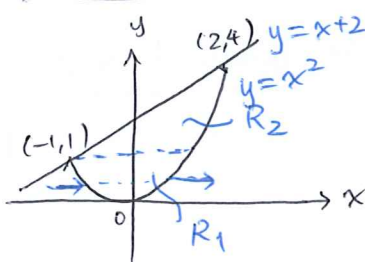
Review of changing limits in integrationFind the area of the region R enclosed by $y = x^2$ and $y = x + 2$.

Solving $\begin{cases} y = x^2 \\ y = x + 2 \end{cases}$

we have $x^2 - x - 2 = 0$
 $(x-2)(x+1) = 0$
 $x = 2 \text{ or } -1$
 $y = 4 \text{ or } 1$

Method 1: Go in y-direction first, then x-direction

$$\int_{-1}^2 \int_{x^2}^{x+2} dy dx$$

Method 2: Go in x-direction first, then y-direction

$$\begin{aligned} \text{Area} &= \iint_{R_1} dA + \iint_{R_2} dA \\ &= \int_0^1 \int_{-\sqrt{y}}^{\sqrt{y}} dx dy \\ &\quad + \int_1^4 \int_{y-2}^{\sqrt{y}} dx dy \end{aligned}$$

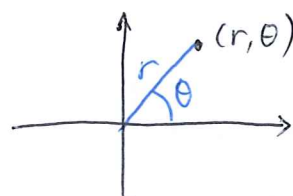
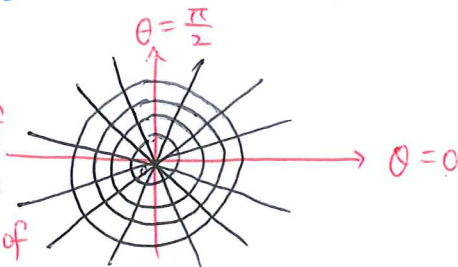
We use Method 1 to carry out the evaluation:

$$\text{Area} = \int_{-1}^2 \left[y \right]_{x^2}^{x+2} dx = \int_{-1}^2 (x+2-x^2) dx = \left[\frac{x^2}{2} + 2x - \frac{x^3}{3} \right]_{-1}^2 = \underline{\underline{\frac{9}{2}}}$$

Polar Coordinate System

In polar coordinate system, we adopt a radial and circular net to describe points and curves on the $\mathbb{R}^2$ plane. Points are called "polar points" and curves are called "polar curves". The origin is called the "pole", while the positive x-axis is called the "polar axis".

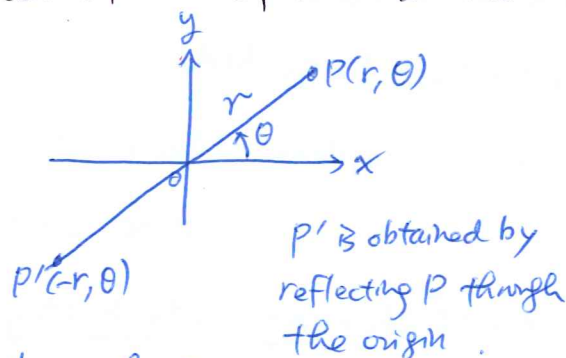
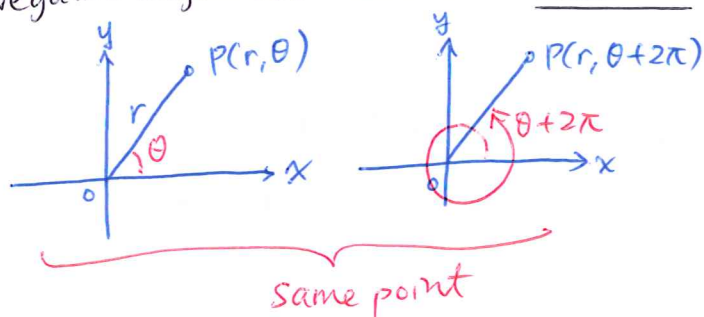
In this chapter, the radial coord. of P describes the signed distance from origin to P, while the angular coord. of P describes angle whose initial side is the positive x-axis and terminal side lies on the ray that passes through O and P.



r : radial coordinate
 θ : angular coordinate

Positive angles are measured in anti-clockwise direction from the positive x-axis. P.2

Negative angles are measured in clockwise direction from the positive x-axis

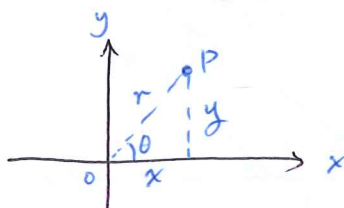


Remarks: In polar coordinate system, each point can have infinitely many representations. Angles are determined up to multiples of 2π rad. r can also be negative.

Conversion of Coordinates

(x, y) : rectangular point

(r, θ) : polar point



Given x and y

Find r and θ by

$$r = \sqrt{x^2 + y^2}$$

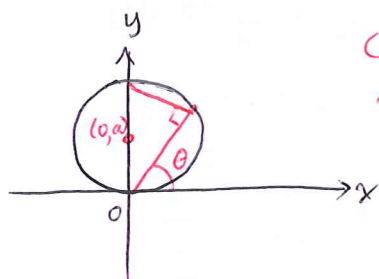
$$\tan \theta = \frac{y}{x}$$

Here we assume $r \geq 0$ for simplicity.

$$\Leftrightarrow \begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases}$$

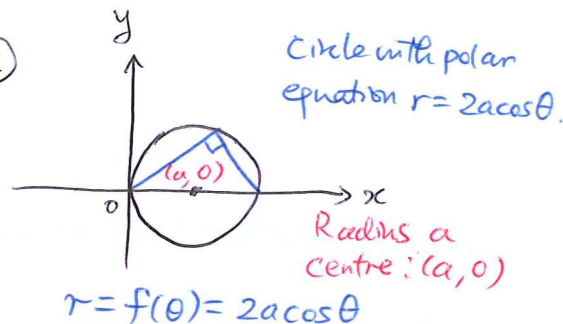
Some examples of curves in Polar coordinate system

① $r = f(\theta) = 2a \sin \theta$



Circle with polar equation $r = 2a \sin \theta$, radius a , centre: $(0, a)$

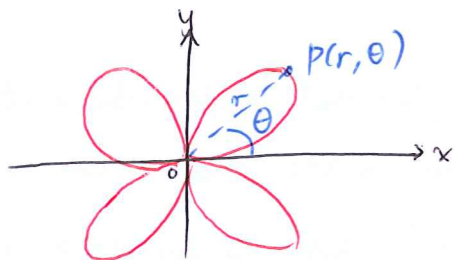
②



Circle with polar equation $r = 2a \cos \theta$

Radius a , centre: $(a, 0)$

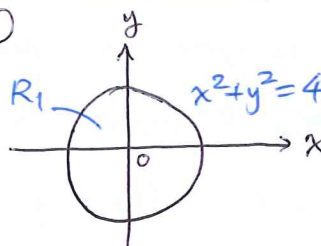
③ $r = f(\theta) = 3 \sin 2\theta$



Four-leaf rose

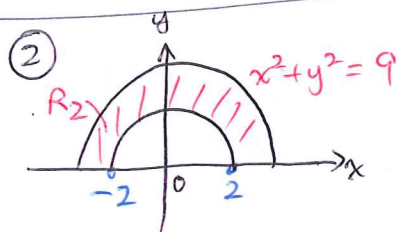
Describing regions in polar coordinates

①

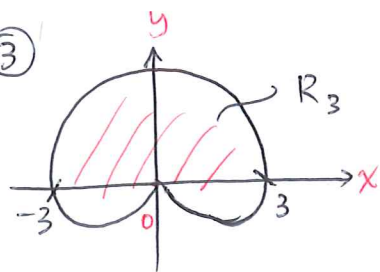


$$R_1 = \{(r, \theta) \mid 0 \leq r \leq 2, 0 \leq \theta \leq 2\pi\}$$

$$R_2 = \{(r, \theta) \mid 2 \leq r \leq 3, 0 \leq \theta \leq \pi\}$$



③



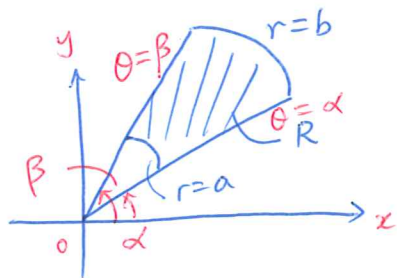
The region R_3 is a cardioid (心臟線) with $r = f(\theta) = 3 + 3\sin\theta$.

$$R_3 = \{(r, \theta) \mid 0 \leq r \leq 3 + 3\sin\theta, 0 \leq \theta \leq 2\pi\}$$

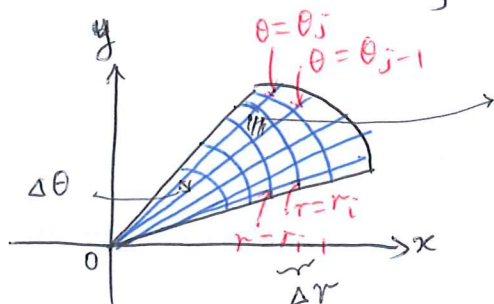
p.3

In order to compute the double integral $\iint_R f(x, y) dA$, where R is the polar rectangle, we divide the interval $[a, b]$ into m sub-intervals $[r_{i-1}, r_i]$ of equal width

$\Delta r = \frac{b-a}{m}$, also we divide the interval $[\alpha, \beta]$ into n sub-intervals $[\theta_{j-1}, \theta_j]$ of equal width $\Delta\theta = \frac{\beta-\alpha}{n}$.



Then we have the following diagram:



$$R_{ij} = \{(r, \theta) \mid r_{i-1} \leq r \leq r_i, \theta_{j-1} \leq \theta \leq \theta_j\}$$

Division of R into polar subrectangles

Consider the area of a wedge-shaped sector of a circle having radius r and angle θ

$$= \pi r^2 \cdot \frac{\theta}{2\pi} = \frac{1}{2} r^2 \theta$$

Area of circle $\times$ fraction of the circle's area contained in the wedge.

$$r_i^* = \frac{1}{2}(r_{i-1} + r_i)$$

$$\theta_j^* = \frac{1}{2}(\theta_{j-1} + \theta_j)$$

$$\therefore \text{Area of } R_{ij} = \frac{1}{2} r_i^2 \Delta\theta - \frac{1}{2} r_{i-1}^2 \Delta\theta, \text{ where } \Delta\theta = \theta_j - \theta_{j-1}$$

$$= \frac{1}{2} (r_i^2 - r_{i-1}^2) \Delta\theta$$

$$= \frac{1}{2} (r_i + r_{i-1})(r_i - r_{i-1}) \Delta\theta$$

$$= r_i^* \Delta r \Delta\theta$$

for discrete sense (in terms of ordinary rectangles)

It can be shown that for continuous functions f , we always obtain the same expression using polar rectangles.

Write $g(r, \theta) = r f(r \cos\theta, r \sin\theta)$,

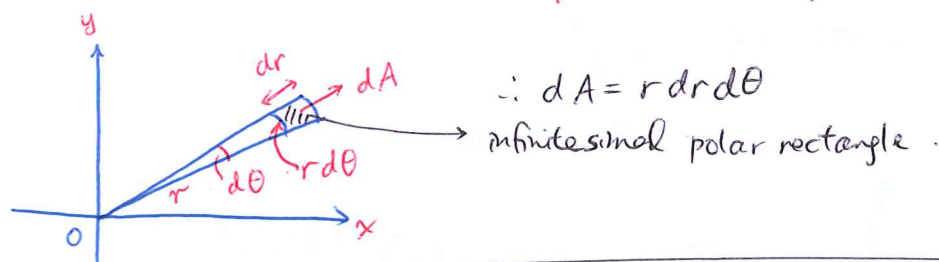
then we have $\sum_{i=1}^m \sum_{j=1}^n g(r_i^*, \theta_j^*) \Delta r \Delta\theta$, which is a Riemann sum for $\int_{\alpha}^{\beta} \int_a^b g(r, \theta) dr d\theta$

$\therefore$ We have $\boxed{\iint_R f(x,y) dA = \lim_{\substack{m \rightarrow \infty \\ n \rightarrow \infty}} \sum_{i=1}^m \sum_{j=1}^n f(r_i^* \cos \theta_j^*, r_i^* \sin \theta_j^*) \Delta A_i}$

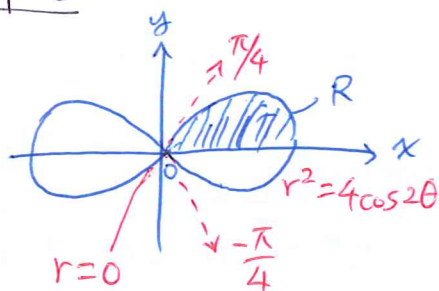
$$= \lim_{\substack{m \rightarrow \infty \\ n \rightarrow \infty}} \sum_{i=1}^m \sum_{j=1}^n g(r_i^*, \theta_j^*) \Delta r \Delta \theta = \int_{\alpha}^{\beta} \int_a^b g(r, \theta) dr d\theta$$

$$= \boxed{\int_{\alpha}^{\beta} \int_a^b f(r \cos \theta, r \sin \theta) r dr d\theta}$$

for continuous function f on a polar rectangle R defined by
 $0 \leq a \leq r \leq b$, $\alpha \leq \theta \leq \beta$, where $0 \leq \beta - \alpha \leq 2\pi$.



Examples ① Find the area enclosed by $r^2 = 4 \cos 2\theta$

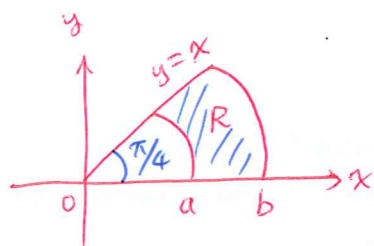


We consider one quarter of the lemniscate.

$$\begin{aligned} \text{Area of } R &= \int_0^{\pi/4} \int_0^{\sqrt{4 \cos 2\theta}} r dr d\theta \\ &= \int_0^{\pi/4} \left[\frac{r^2}{2} \right]_0^{\sqrt{4 \cos 2\theta}} d\theta \\ &= \int_0^{\pi/4} 2 \cos 2\theta d\theta = \left[\sin 2\theta \right]_0^{\pi/4} \\ &= 1 \end{aligned}$$

$\Rightarrow$ Area of the lemniscate $= 4 \times 1 = 4$ (by symmetry)

② (P. 835 of Adams and Essex) If R is the part of annulus $0 \leq a^2 \leq x^2 + y^2 \leq b^2$ lying in the first quadrant and below $y = x$, evaluate $\iint_R \frac{y^2}{x^2} dA$



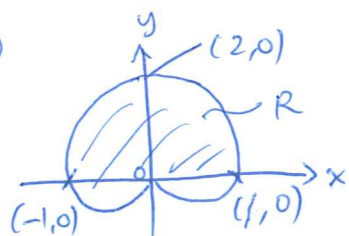
Consider $\frac{y^2}{x^2} = \frac{r^2 \sin^2 \theta}{r^2 \cos^2 \theta} = \tan^2 \theta$ in polar coordinates.

$$R = \{(r, \theta) \mid a \leq r \leq b, 0 \leq \theta \leq \frac{\pi}{4}\}$$

$$\begin{aligned} \therefore \iint_R \frac{y^2}{x^2} dA &= \int_0^{\pi/4} \int_a^b \tan^2 \theta r dr d\theta \\ &= \int_0^{\pi/4} \tan^2 \theta d\theta \int_a^b r dr \\ &= \frac{1}{2} (b^2 - a^2) \int_0^{\pi/4} (\sec^2 \theta - 1) d\theta \\ &= \frac{1}{2} (b^2 - a^2) (\tan \theta - \theta) \Big|_0^{\pi/4} = \frac{1}{2} (b^2 - a^2) \left(1 - \frac{\pi}{4}\right) \dots \end{aligned}$$

③

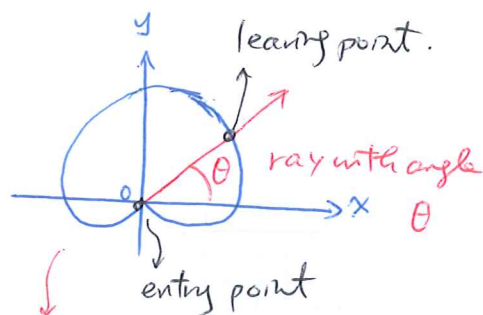
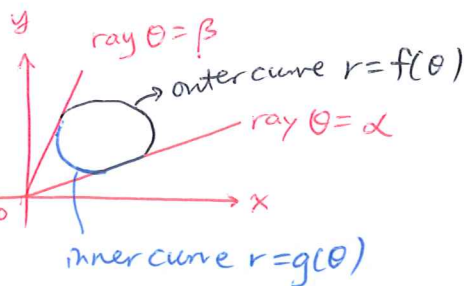
P. 5



Given $r = f(\theta) = 1 + \sin \theta$.

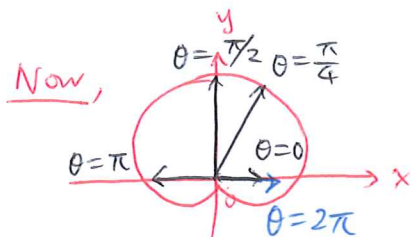
Find the area of the region enclosed (R).

$$\int_{\theta=\alpha}^{\theta=\beta} \int_{r=g(\theta)}^{r=f(\theta)} r dr d\theta$$



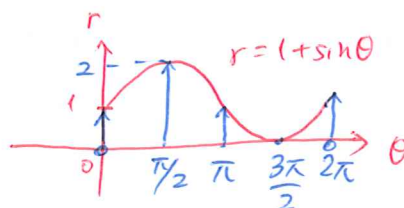
$$r_{\text{inner}} = g(\theta) = 0$$

$$r_{\text{outer}} = f(\theta) = 1 + \sin \theta$$



starting ray: $\theta = 0$, i.e. $\alpha = 0$

ending ray: $\theta = 2\pi$, i.e. $\beta = 2\pi$



$$\begin{aligned} \therefore \text{Area} &= \int_0^{2\pi} \frac{1}{2} (f(\theta)^2 - g(\theta)^2) d\theta = \frac{1}{2} \int_0^{2\pi} (1 + \sin \theta)^2 d\theta \\ &= \int_0^{2\pi} \left[\frac{1}{2} + \sin \theta + \frac{1 - \cos 2\theta}{4} \right] d\theta \\ &= \left[\frac{\theta}{2} - \cos \theta + \frac{\theta}{4} - \frac{\sin 2\theta}{8} \right]_0^{2\pi} = \frac{3\pi}{2} \end{aligned}$$

④ Evaluate $\int_1^2 \int_0^{\sqrt{2x-x^2}} y dy dx$ using polar coordinates.

$$R = \{(x, y) \mid 1 \leq x \leq 2 \text{ and } 0 \leq y \leq \sqrt{2x-x^2}\}$$

For $x = 1$, $r \cos \theta = 1 \Leftrightarrow r = \sec \theta$

For $y = \sqrt{2x-x^2}$, $r \sin \theta = \sqrt{2r \cos \theta - r^2 \cos^2 \theta}$

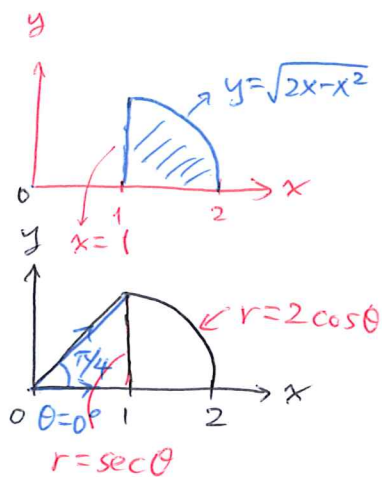
$$r^2 \sin^2 \theta = 2r \cos \theta - r^2 \cos^2 \theta$$

$$r = 2 \cos \theta$$

$$\therefore \int_1^2 \int_0^{\sqrt{2x-x^2}} y dy dx = \int_{\pi/4}^{\pi/2} \int_{\sec \theta}^{2 \cos \theta} (r \sin \theta) r dr d\theta$$

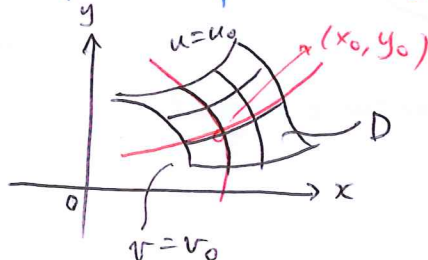
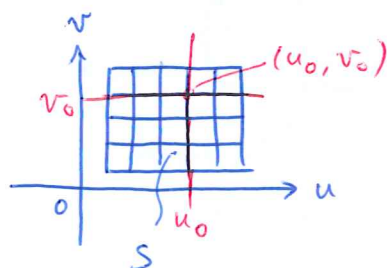
$$= \int_{\pi/4}^{\pi/2} \int_{\sec \theta}^{2 \cos \theta} r^2 \sin \theta dr d\theta$$

$$= \int_{\pi/4}^{\pi/2} \sin \theta \left. \frac{1}{3} r^3 \right|_{\sec \theta}^{2 \cos \theta} d\theta : \dots$$



The transformation of a double integral to polar coordinates is a special case of a general change of variables formula for double integrals. P.6

Suppose $x = x(u, v)$
 $y = y(u, v)$ (i.e. transformation from points (u, v) in a uv -Cartesian plane to points (x, y) in the xy -plane)



Under the transformation, the lines $u = u_0$ and $v = v_0$ in the uv -plane are mapped to the curves $\begin{cases} x = x(u_0, v) \\ y = y(u_0, v) \end{cases}$ and $\begin{cases} x = x(u, v_0) \\ y = y(u, v_0) \end{cases}$ in the xy -plane.

i.e. $(u_0, v_0) \mapsto (x_0, y_0)$

The transformation from the set S in uv -plane to the set D in xy -plane is bijective if

- ① every point in S gets mapped to a point in D
- ② every point in D is the image of a point in S
- ③ different points in S are mapped to different points in D .

If the transformation is one-to-one, we can write $u = u(x, y)$, $v = v(x, y)$.

Assume the functions $x(u, v)$ and $y(u, v)$ are C^1 and $\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix} \neq 0$

at (u, v) , then the transformation is one-to-one near (u, v) and the inverse

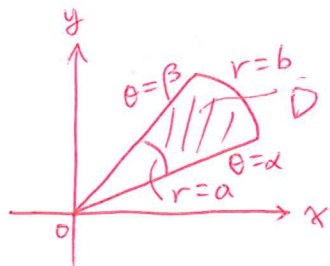
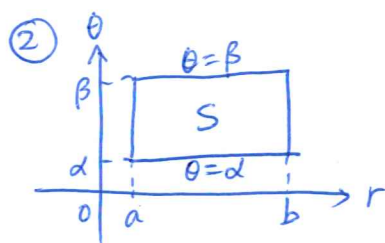
transformation $\frac{\partial(u, v)}{\partial(x, y)}$ also has continuous first partial derivatives and non-zero

Jacobian such that $\frac{\partial(u, v)}{\partial(x, y)} = \frac{1}{\frac{\partial(x, y)}{\partial(u, v)}}$ on the set D .

Example

① In particular, $\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases}$ has Jacobian $\frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r$

Hence, near any point except the origin ($r = 0$), the transformation is one-to-one.



$$\begin{aligned} \iint_D f(x, y) dA &= \iint_S f(r \cos \theta, r \sin \theta) r dr d\theta \\ &= \iint_S f(r \cos \theta, r \sin \theta) \left| \frac{\partial(x, y)}{\partial(r, \theta)} \right| dr d\theta \quad (r > 0) \end{aligned}$$

Theorem (Change of variables)

P.7

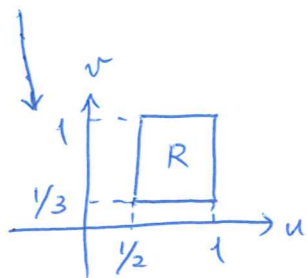
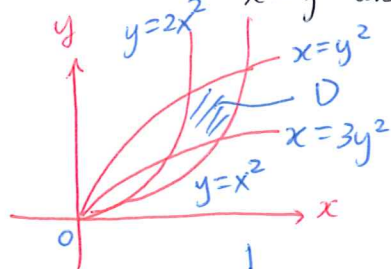
Let $x = x(u, v)$, $y = y(u, v)$ be a one-to-one transformation from a domain S in the uv -plane onto a domain D in the xy -plane.

Suppose x and y and their first partial derivatives with respect to u and v are continuous in S , if $f(x, y)$ is integrable on D , $g(u, v) = f(x(u, v), y(u, v))$,

then g is integrable on S and $\iint_D f(x, y) dx dy = \iint_S g(u, v) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du dv$

Example ① Find the area of the finite plane region bounded by $y = x^2$, $y = 2x^2$,

$x = y^2$ and $x = 3y^2$.



Let $u = \frac{x^2}{y}$, $v = \frac{y^2}{x}$, then the region D in xy -plane is mapped to the region R in uv -plane, given by

$$\{(u, v) \mid \frac{1}{2} \leq u \leq 1 \text{ and } \frac{1}{3} \leq v \leq 1\}$$

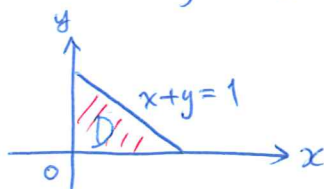
$$\text{Consider } \frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \frac{2x}{y} & -\frac{x^2}{y^2} \\ -\frac{y^2}{x^2} & \frac{2y}{x} \end{vmatrix} = 3$$

$$\Rightarrow \left| \frac{\partial(x, y)}{\partial(u, v)} \right| = \frac{1}{3}$$

$$\therefore \text{Area} = \iint_D dx dy = \iint_R \frac{1}{3} du dv = \frac{1}{3} \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{3}\right) = \frac{1}{9}$$

② Evaluate $\int_0^1 \int_0^{1-x} \sqrt{x+y} (y-2x)^2 dy dx$

Sketch region D



Integrand suggests we substitute

$u = x+y$ and $v = y-2x$.

$$\text{Solving yields } \begin{cases} x = \frac{u}{3} - \frac{v}{3} \\ y = \frac{2u}{3} + \frac{v}{3} \end{cases}$$

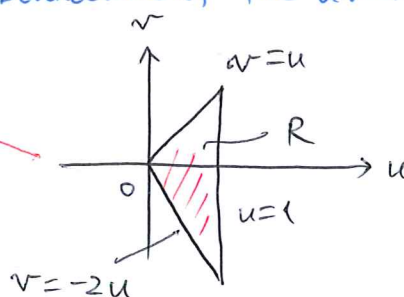
Note: The equations

$$\begin{cases} x = \frac{u}{3} - \frac{v}{3} \\ y = \frac{2u}{3} + \frac{v}{3} \end{cases} \text{ transform}$$

R to D .

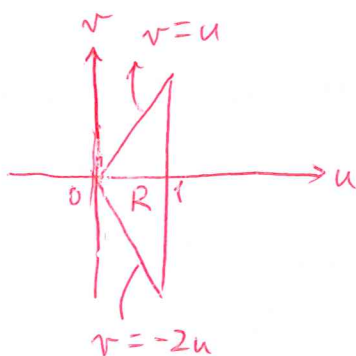
$$\begin{cases} u = x+y \\ v = y-2x \end{cases} \text{ transforms } D \text{ to } R.$$

Then we can try figure out the boundaries of the uv -region R .

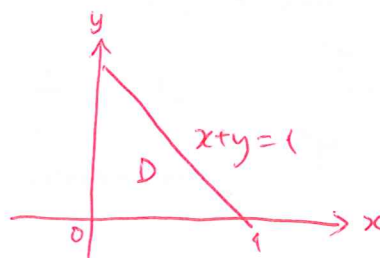


We construct a table as follows:

xy-equations for boundary of D	uv-equations for boundary of R	Simplified uv-equations
$x+y=1$	$(\frac{u}{3}-\frac{v}{3})+(\frac{2u}{3}+\frac{v}{3})=1$	$u=1$
$x=0$	$\frac{u}{3}-\frac{v}{3}=0$	$v=u$
$y=0$	$\frac{2u}{3}+\frac{v}{3}=0$	$v=-2u$



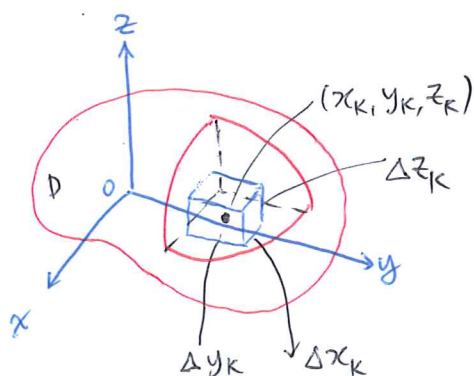
$$\begin{cases} x = \frac{u}{3} - \frac{v}{3} \\ y = \frac{2u}{3} + \frac{v}{3} \end{cases}$$



$$\text{Jacobian} = \begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix} = \begin{vmatrix} \frac{1}{3} & -\frac{1}{3} \\ \frac{2}{3} & \frac{1}{3} \end{vmatrix} = \frac{1}{3}$$

$$\begin{aligned} \Rightarrow \int_0^1 \int_0^{1-x} \sqrt{x+y} (y-2x)^2 dy dx &= \int_0^1 \int_{-2u}^u u^{\frac{1}{2}} v^2 \left(\frac{1}{3} \right) dv du \\ &= \frac{1}{3} \int_0^1 u^{\frac{1}{2}} \left[\frac{1}{3} v^3 \right]_{-2u}^u du \\ &= \int_0^1 u^{\frac{7}{2}} du = \frac{2}{9} u^{\frac{9}{2}} \Big|_0^1 = \underline{\underline{\frac{2}{9}}} \end{aligned}$$

Triple Integration



$F(x, y, z)$: a function defined on a closed and bounded region D in space, such as the region occupied by a solid ball or a lump of clay.

We partition a rectangular boxlike region containing D into rectangular cells by planes parallel to x, y and z axes.

Consider the k th cell with dimensions Δx_k by Δy_k by Δz_k .

$$\text{i.e. } \Delta V_k = \Delta x_k \Delta y_k \Delta z_k.$$

Select a point (x_k, y_k, z_k) in each cell, let $S_n = \sum_{k=1}^n F(x_k, y_k, z_k) \Delta V_k$.

Suppose $\Delta x_k, \Delta y_k, \Delta z_k$ and $\max \{\Delta x_k, \Delta y_k, \Delta z_k\} \rightarrow 0$, i.e. as $\|P\| \rightarrow 0$ and

the no. of cells $n \rightarrow \infty$, the sums S_n approaches a limit. Such limit is the "triple integral of F over D ",

$$\lim_{n \rightarrow \infty} S_n = \iiint_D F(x, y, z) dV$$

$$\text{or } \lim_{\|P\| \rightarrow 0} S_n = \iiint_D F(x, y, z) dx dy dz$$

(provided that limit exists)

region over which continuous functions are integrable, i.e. can be closely approximated by small rectangular cells.

Definition The volume of a closed and bounded region D in 3-space is defined as

$$\iiint_D 1 dV \stackrel{\text{def}}{=} \iiint_D dV.$$

Evaluation by Iterated Integrals

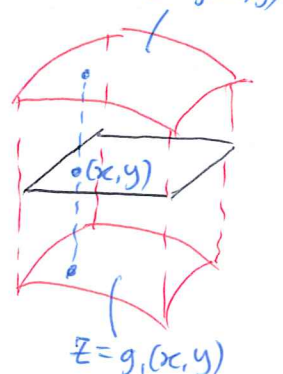
Let f be a continuous function on a solid region D defined by

$$a \leq x \leq b$$

$$h_1(x) \leq y \leq h_2(x), \text{ where } h_1, h_2, g_1, \text{ and } g_2 \text{ are continuous functions.}$$

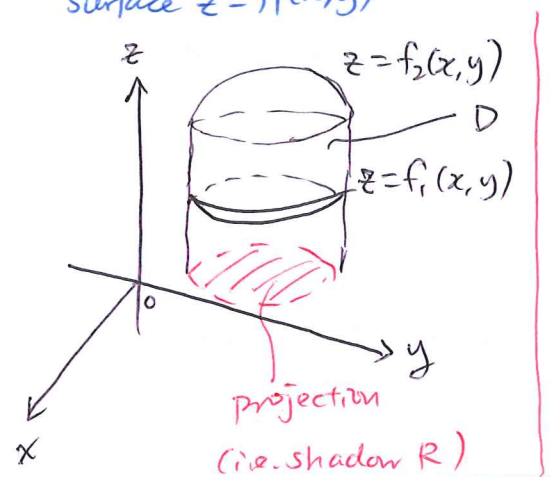
$$g_1(x, y) \leq z \leq g_2(x, y)$$

$$\text{Then } \iiint_D f(x, y, z) dV = \int_a^b \int_{h_1(x)}^{h_2(x)} \int_{g_1(x, y)}^{g_2(x, y)} f(x, y, z) dz dy dx$$



Procedures for finding lower and upper limits of a triple integral over a closed solid region D bounded above by surface $z = f_2(x, y)$, and bounded below by surface $z = f_1(x, y)$

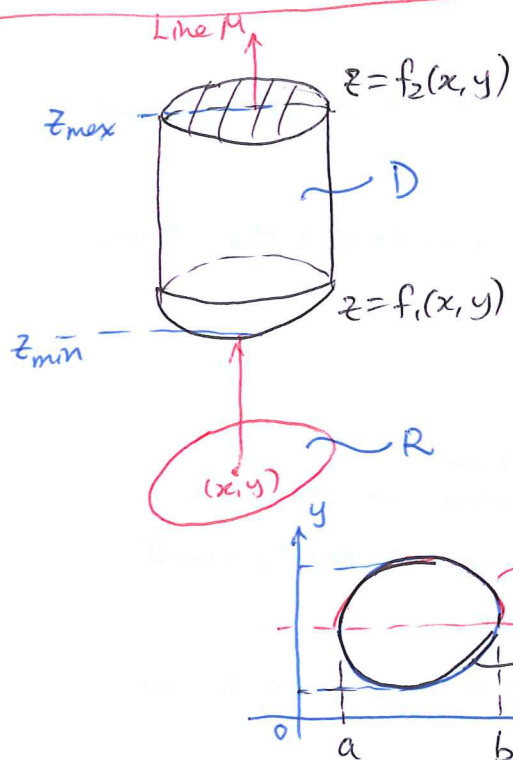
P.10



$$\iiint_D F(x, y, z) dV = \int_{x=?}^{x=?} \int_{y=?}^{y=?} \int_{z=?}^{z=?} F(x, y, z) dz dy dx$$

innermost integrals.

① Look at the shadow of the solid region D onto the xy -plane. Fix a point (x, y) within the shadow R . Then at that point, we draw a line M that passes through the solid region D parallel to the z -axis in direction of increasing z .



② After evaluating the innermost integral, no more variable z . Let the innermost integral be $G(x, y)$.

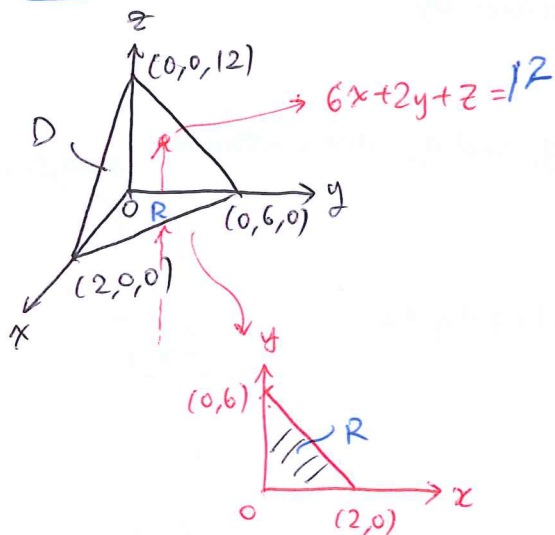
$$\iiint_D F(x, y, z) dV = \iint_R G(x, y) dy dx$$

vertical shadow of the solid region D in the xy -plane

$$= \int_a^b \int_{g_1(x)}^{g_2(x)} G(x, y) dy dx$$

$$\int_a^b \int_{g_1(x)}^{g_2(x)} \int_{f_1(x, y)}^{f_2(x, y)} F(x, y, z) dz dy dx$$

Examples ① Find the volume of the following bounded region D in the 1st octant.



$$\text{Volume} = \iiint_D 1 \cdot dV = \iint_R (12 - 6x - 2y) dA_R$$

$$= \int_0^2 \int_0^{6-3x} (12 - 6x - 2y) dy dx$$

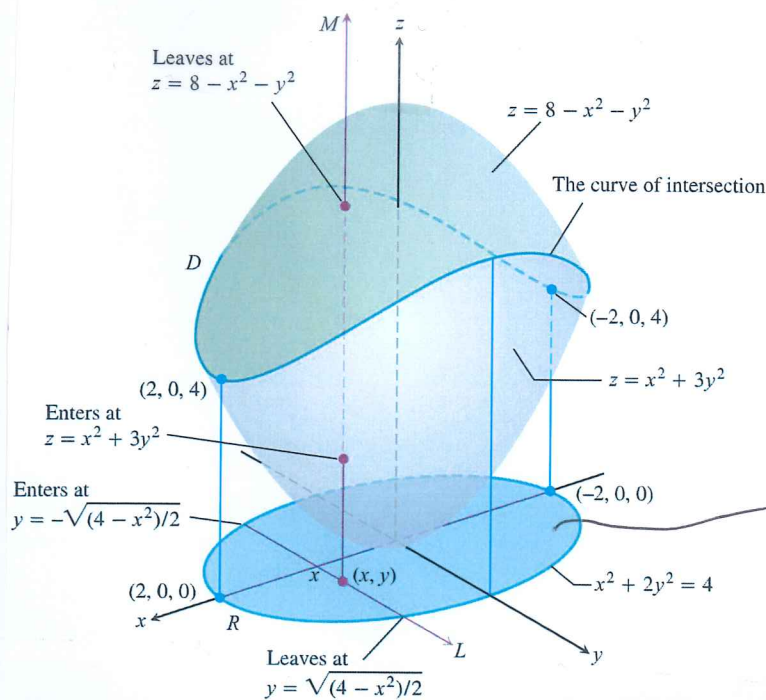
$$= \int_0^2 \int_0^{6-3x} (12 - 6x - 2y) dy dx$$

$$= 24$$

② (Thomas Calculus
P.1101-1102)

Find the volume of the region D enclosed by the surfaces
 $z = x^2 + 3y^2$ and $z = 8 - x^2 - y^2$.

P.11



$$\begin{aligned}\text{Volume of } D &= \iiint_D 1 \, dV \\ &= \iint_R \int_{z=x^2+3y^2}^{z=8-x^2-y^2} 1 \, dz \, dy \, dx \\ &= \iint_R (8-2x^2-4y^2) \, dy \, dx\end{aligned}$$

We need to find the
equation of the boundary
curve of shadow R.

Shadow
of the solid
region D onto
xy-plane.

Consider $x^2 + 3y^2 = 8 - x^2 - y^2$
(the place where they intersect
together must have the same
z-value)

$$\text{i.e. } x^2 + 2y^2 = 4 \text{ (ellipse)}$$

The line L enters into region R at $y = -\sqrt{\frac{4-x^2}{2}}$ and leaves region R at $y = \sqrt{\frac{4-x^2}{2}}$.

$$\therefore \text{Volume of } D = \int_{-2}^2 \int_{y=-\sqrt{\frac{4-x^2}{2}}}^{y=\sqrt{\frac{4-x^2}{2}}} (8-2x^2-4y^2) \, dy \, dx$$

$$= \int_{-2}^2 \left(2(8-2x^2) \sqrt{\frac{4-x^2}{2}} - \frac{8}{3} \left(\frac{4-x^2}{2} \right)^{\frac{3}{2}} \right) dx$$

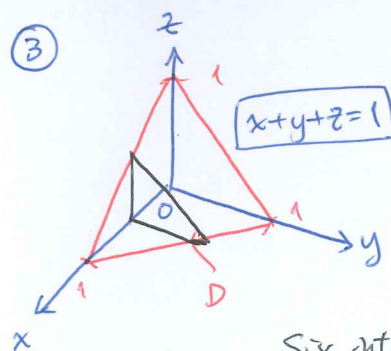
$$= \int_{-2}^2 \left[8 \left(\frac{4-x^2}{2} \right)^{\frac{3}{2}} - \frac{8}{3} \left(\frac{4-x^2}{2} \right)^{\frac{3}{2}} \right] dx$$

$$= \frac{4\sqrt{2}}{3} \int_{-2}^2 (4-x^2)^{\frac{3}{2}} dx$$

$$= \underline{\underline{8\sqrt{2}\pi}}$$

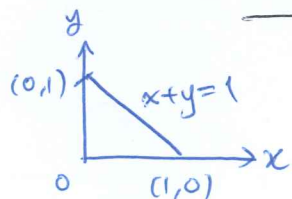
$$\begin{aligned}\text{Let } x &= 2 \sin u \\ dx &= 2 \cos u \, du\end{aligned}$$

③



Let D be the tetrahedron with vertices $(0,0,0)$, $(1,0,0)$, $(0,1,0)$ and $(0,0,1)$. Write out the 6 integrals for $\iiint_D y \, dV$, and evaluate this triple integral via any order.

P.12



Six integration orders

$$\int_0^1 \int_0^{1-x} \int_0^{1-x-y} y \, dz \, dy \, dx$$

$$\int_0^1 \int_0^{1-y} \int_0^{1-x-y} y \, dz \, dx \, dy$$

$$\int_0^1 \int_0^{1-x} \int_0^{1-x-z} y \, dy \, dz \, dx$$

$$\int_0^1 \int_0^{1-z} \int_0^{1-x-z} y \, dy \, dx \, dz$$

$$\int_0^1 \int_0^{1-z} \int_0^{1-y-z} y \, dx \, dy \, dz$$

$$\int_0^1 \int_0^{1-y} \int_0^{1-y-z} y \, dx \, dz \, dy$$

$$\begin{aligned} &= \int_0^1 \int_0^{1-x} y(1-x-y) \, dy \, dx \\ &= \int_0^1 \left[(1-x) \frac{y^2}{2} - \frac{y^3}{3} \right]_0^{1-x} dx \\ &= \int_0^1 \frac{1}{6} (1-x)^3 dx = \underline{\underline{\frac{1}{24}}} \end{aligned}$$

Applications

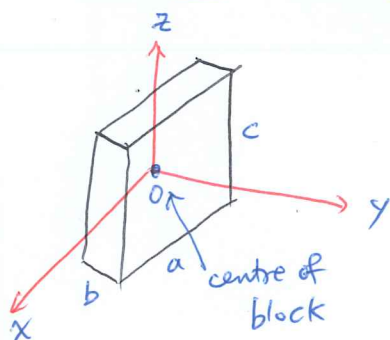
Mass of solid objects in space $M = \iiint_D \delta \, dV$ ($\delta = \delta(x,y,z) = \text{density}$)

Centre of mass $(\bar{x}, \bar{y}, \bar{z}) = \left(\frac{\iiint_D x \delta \, dV}{M}, \frac{\iiint_D y \delta \, dV}{M}, \frac{\iiint_D z \delta \, dV}{M} \right)$ first moments about coord. planes

Moment of Inertia about the coordinate axes:

$$I_x = \iiint (y^2 + z^2) \delta \, dV; \quad I_y = \iiint (x^2 + z^2) \delta \, dV; \quad I_z = \iiint (x^2 + y^2) \delta \, dV.$$

Example Find I_x , I_y and I_z for the rectangular solid with constant density δ .



M : mass of the block.

$$I_x = \int_{-c/2}^{c/2} \int_{-b/2}^{b/2} \int_{-a/2}^{a/2} (y^2 + z^2) \delta \, dx \, dy \, dz$$

even function of x, y, z .

Totally 8 symmetric pieces, one in

$$= 8 \int_0^{c/2} \int_0^{b/2} \int_0^{a/2} (y^2 + z^2) \delta \, dx \, dy \, dz$$

$$= 4a\delta \int_0^{c/2} \int_0^{b/2} (y^2 + z^2) \, dy \, dz$$

$$= 4a\delta \int_0^{c/2} \left[\frac{y^3}{3} + z^2 y \right]_0^{b/2} dz$$

$$= 4a\delta \int_0^{c/2} \left(\frac{b^3}{24} + \frac{z^2 b}{2} \right) dz = \frac{abc\delta}{12} (b^2 + c^2) = \underline{\underline{\frac{M}{12} (b^2 + c^2)}}$$

By symmetry,

$$I_y = \frac{M}{12} (a^2 + c^2)$$

$$I_z = \frac{M}{12} (a^2 + b^2)$$

Triple Integrals in Cylindrical Coordinates and Spherical Coordinates

P.13

The change of variables formula in double integral can be extended to triple (or higher order) integrals.

Consider $\begin{cases} x = x(u, v, w) \\ y = y(u, v, w) \\ z = z(u, v, w) \end{cases}$, where x, y and z have continuous first partial derivatives

with respect to u, v, w . Then near any point where $\frac{\partial(x, y, z)}{\partial(u, v, w)} \neq 0$,

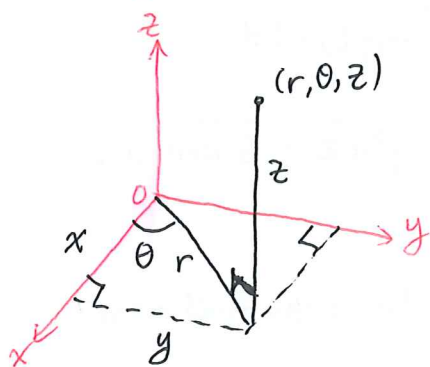
$$dV = dx dy dz = \left| \frac{\partial(x, y, z)}{\partial(u, v, w)} \right| du dv dw$$

If the transformation is one-to-one from a domain S in uvw -space onto a domain D in xyz -space, if $g(u, v, w) = f(x(u, v, w), y(u, v, w), z(u, v, w))$, then

$$\iiint_D f(x, y, z) dx dy dz = \iiint_S g(u, v, w) \left| \frac{\partial(x, y, z)}{\partial(u, v, w)} \right| du dv dw.$$

Integration in Cylindrical Coordinates

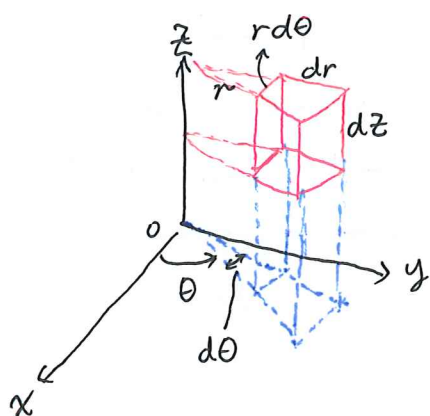
Def: Cylindrical coordinates represent a point P in space defined by ordered triples (r, θ, z) , where we assume $r \geq 0$ in our following context.



$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \\ z = z \end{cases} \quad \begin{aligned} r^2 &= x^2 + y^2 \\ \tan \theta &= \frac{y}{x} \end{aligned}$$

Volume element in cylindrical coordinates: $dV = r dr d\theta dz$

We examine the infinitesimal 'box' bounded by the coordinate surfaces corresponding to values $r, r+dr, \theta, \theta+d\theta, z$ and $z+dz$.



$$\frac{\partial(x, y, z)}{\partial(r, \theta, z)} = \begin{vmatrix} \cos \theta & -r \sin \theta & 0 \\ \sin \theta & r \cos \theta & 0 \\ 0 & 0 & 1 \end{vmatrix} = r$$

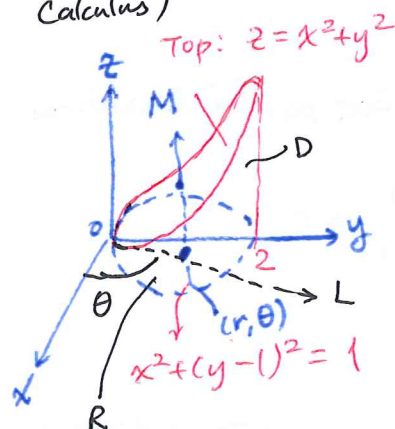
$$\therefore \iiint_D f(x, y, z) dx dy dz = \iiint_D f(r, \theta, z) r dr d\theta dz$$

$$\theta: \alpha \leq \theta \leq \beta, \text{ where } 0 \leq \beta - \alpha \leq 2\pi$$

Examples ① Find the limits of integration in cylindrical coordinates for integrating P.14

(due to Thomas Calculus)

a function $f(r, \theta, z)$ over the region D bounded by $z=0$, laterally by the circular cylinder $x^2 + (y-1)^2 = 1$, and above by paraboloid $z = x^2 + y^2$.



Project solid D onto xy -plane, the projection is denoted as R . The boundary of R is the circle $x^2 + (y-1)^2 = 1$.

Consider its polar coord. equation: $r^2 - 2r \sin \theta = 0$
 $r = 2 \sin \theta$.

z -limits of integration

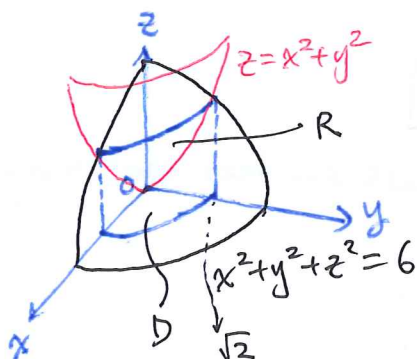
Consider a line M through a typical point (r, θ) in R parallel to the z -axis: It enters D at $z=0$ and leaves at $z = x^2 + y^2$.

r -limits of integration: A ray L through (r, θ) from origin enters R at $r=0$ and leaves R at $r = 2 \sin \theta$.

θ -limits of integration: As L sweeps across R , the angle θ it makes with the positive x -axis run from $\theta=0$ to $\theta=\pi$.

$$\iiint_D f(r, \theta, z) dv = \int_0^\pi \int_0^{2 \sin \theta} \int_0^{x^2 + y^2} f(r, \theta, z) r dz dr d\theta$$

② Find the volume of the solid region inside the sphere $x^2 + y^2 + z^2 = 6$ and above the paraboloid $z = x^2 + y^2$.



One quarter of the required volume lies in the first octant.

Denote one quarter of the solid region as R , and its projection onto xy -plane: one quarter of the disk be D .

The two surfaces intersect on the vertical cylinder:

$$6 - x^2 - y^2 = z^2 = (x^2 + y^2)^2, \text{ i.e. } 6 - r^2 = r^4$$

$$(r^2 + 3)(r^2 - 2) = 0$$

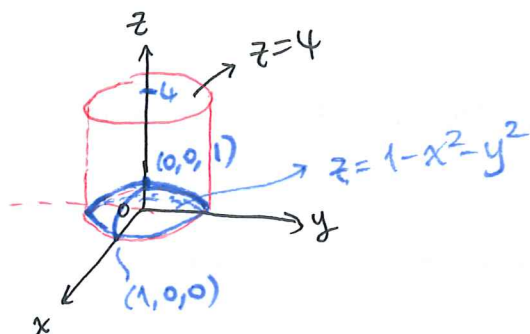
The only relevant solⁿ is $r = \sqrt{2}$, hence the required volume lies above the disk D centered at origin in the xy -plane.

$$\begin{aligned} \therefore V &= \iiint_R dv = \int_0^{2\pi} \int_0^{\sqrt{2}} \int_{r^2}^{\sqrt{6-r^2}} r dz dr d\theta = 2\pi \int_0^{\sqrt{2}} (r\sqrt{6-r^2} - r^3) dr \\ &= 2\pi \left[-\frac{1}{3}(6-r^2)^{\frac{3}{2}} - \frac{r^4}{4} \right]_0^{\sqrt{2}} = \frac{2\pi}{3} (6\sqrt{6} - 11) \end{aligned}$$

③ (mass of an object) A solid G lies within the cylinder $x^2 + y^2 = 1$, bounded below by $z = 4$ and bounded above by the paraboloid $z = 1 - x^2 - y^2$.

The density at any point is proportional to its distance from the axis of the cylinder.

Find the mass of G .



In cylindrical coordinates, the cylinder: $r = 1$

the paraboloid: $z = 1 - r^2$

$$\therefore G = \{(r, \theta, z) \mid 0 \leq \theta \leq 2\pi, 0 \leq r \leq 1, 1 - r^2 \leq z \leq 4\}$$

Density function $f(x, y, z) = k\sqrt{x^2 + y^2} = kr$, where k is a constant.

distance from z -axis

$$\therefore \text{Mass} = \iiint_G k\sqrt{x^2 + y^2} dV = \int_0^{2\pi} \int_0^1 \int_{1-r^2}^4 (kr) r dz dr d\theta$$

$$= \int_0^{2\pi} \int_0^1 kr^2(3 + r^2) dr d\theta$$

$$= 2\pi k \left[r^3 + \frac{r^5}{5} \right]_0^1 = \frac{12\pi k}{5}$$

Integration in Spherical Coordinates

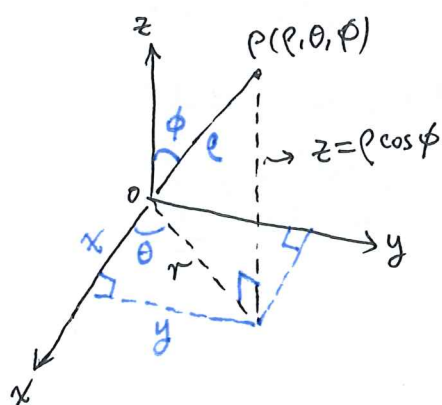
Def.: Spherical coordinates represent a point P in space defined by ordered tuples

(ρ, θ, ϕ) such that

① ρ is the distance from P to the origin ($\rho \geq 0$)

② θ : angle from cylindrical coordinates

③ ϕ : angle $\vec{OP}$ makes with the positive z -axis ($0 \leq \phi \leq \pi$)

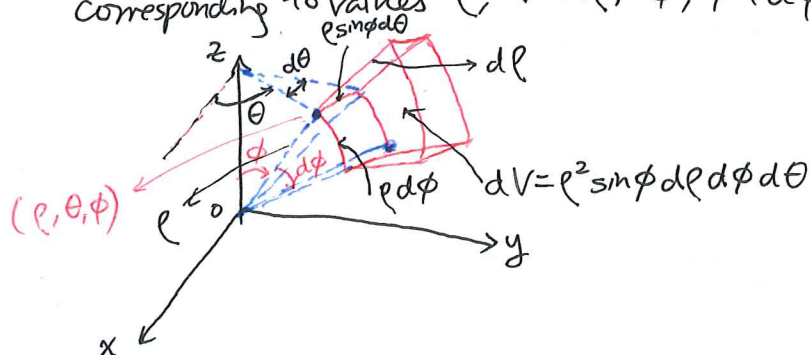


$$\begin{cases} r = \rho \sin \phi & \rightarrow x = r \cos \theta = \rho \sin \phi \cos \theta \\ z = \rho \cos \phi & y = r \sin \theta = \rho \sin \phi \sin \theta \end{cases}$$

$$\rho = \sqrt{x^2 + y^2 + z^2} = \sqrt{r^2 + z^2}$$

Volume element in spherical coordinates: $dV = \rho^2 \sin \phi d\rho d\phi d\theta$

We examine the infinitesimal coordinate box bounded by the coordinate surfaces corresponding to values $\rho, \rho + d\rho, \phi, \phi + d\phi, \theta, \theta + d\theta$.



$$\frac{\partial(x, y, z)}{\partial(\rho, \phi, \theta)} = \begin{vmatrix} \sin \phi \cos \theta & \rho \cos \phi \cos \theta & -\rho \sin \phi \sin \theta \\ \sin \phi \sin \theta & \rho \cos \phi \sin \theta & \rho \sin \phi \cos \theta \\ \cos \phi & -\rho \sin \phi & 0 \end{vmatrix}$$

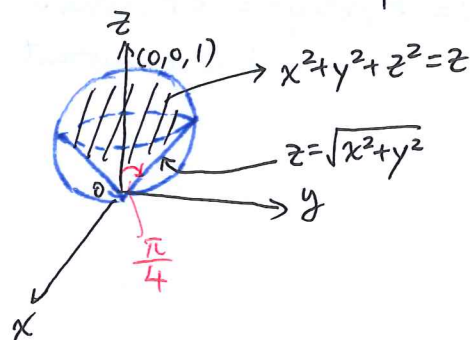
$$= \rho^2 \sin \phi$$

Remarks: Spherical coordinates are suitably designed for problems involving spherical symmetry, e.g., regions bounded by spheres centered at origin, circular cones with axes along the z -axis, and vertical planes containing the z -axis. P.16

$$\iiint_D f(\rho, \phi, \theta) dV = \iiint_D f(\rho, \phi, \theta) \rho^2 \sin \phi d\rho d\phi d\theta$$

$$\alpha \leq \theta \leq \beta, 0 \leq \beta - \alpha \leq 2\pi$$

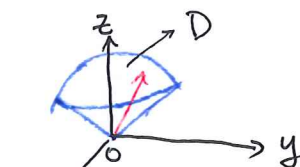
Examples ① Find the volume of the solid that lies above the cone $z = \sqrt{x^2 + y^2}$ and below the sphere $x^2 + y^2 + z^2 = z$.



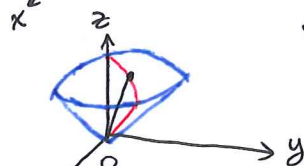
The sphere passes through the origin and has centre $(0, 0, \frac{1}{2})$.

Spherical equation of sphere: $\rho^2 = \rho \cos \phi$
 $\rho = \cos \phi$

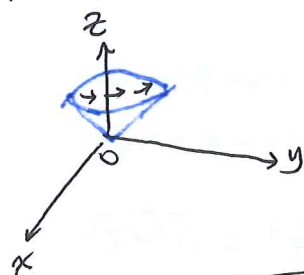
spherical equation of cone: $\rho \cos \phi = \sqrt{\rho^2 \sin^2 \phi \cos^2 \theta + \rho^2 \sin^2 \phi \sin^2 \theta}$
 $= \rho \sin \phi$
 $\phi = \frac{\pi}{4}$



ρ varies from 0 to $\cos \phi$



ϕ varies from 0 to $\frac{\pi}{4}$



θ varies from 0 to 2π

$$\begin{aligned} \therefore \text{Volume} &= \iiint_D dV = \int_0^{2\pi} \int_0^{\frac{\pi}{4}} \int_0^{\cos \phi} \rho^2 \sin \phi d\rho d\phi d\theta \\ &= 2\pi \int_0^{\frac{\pi}{4}} \sin \phi \left[\frac{\rho^3}{3} \right]_0^{\cos \phi} d\phi \\ &= \frac{2\pi}{3} \int_0^{\frac{\pi}{4}} \sin \phi \cos^3 \phi d\phi \\ &= \frac{2\pi}{3} \left[-\frac{\cos^4 \phi}{4} \right]_0^{\frac{\pi}{4}} = \underline{\underline{\frac{\pi}{8}}} \end{aligned}$$

② A solid half-ball B of radius a has density δ depending on distance ρ from the centre of the base disk. Given that $\delta = k(2a - \rho)$, with k being a constant. Find the mass of the half-ball B .

We select the coordinates with origin at the centre of the base (projection), then the half-ball B lies above the xy -plane. Half-ball

$$\begin{aligned} \text{Mass} &= \iiint_B k(2a - \rho) dV = \iiint_B k(2a - \rho) \rho^2 \sin \phi d\rho d\phi d\theta = k \cdot 2\pi \int_0^{\frac{\pi}{2}} \sin \phi d\phi \int_0^a (2a - \rho) \rho^2 d\rho \\ &= 2k\pi \left[\frac{2a}{3} \rho^3 - \frac{\rho^4}{4} \right]_0^a = \underline{\underline{\frac{5}{6} \pi k a^4}} \end{aligned}$$